

Finance & Economics: What They Expect You to Know

Micro, Macro, Accounting, Corporate Finance, Investments, Quant Finance, and Game Theory

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Part I

Microeconomics

Chapter 1

Consumer Theory

1.1 Preferences and Utility

Definition 1.1.1: Preference Relation

A preference relation $\succeq$ on a consumption set $X \subseteq \mathbb{R}_+^n$ satisfies:

- **Completeness:** for all $x, y \in X$, either $x \succeq y$ or $y \succeq x$ (or both).
- **Transitivity:** if $x \succeq y$ and $y \succeq z$, then $x \succeq z$.

We write $x \succ y$ (strict preference) if $x \succeq y$ but not $y \succeq x$, and $x \sim y$ (indifference) if both $x \succeq y$ and $y \succeq x$.

Definition 1.1.2: Utility Function

A function $u : X \rightarrow \mathbb{R}$ **represents** the preference relation $\succeq$ if for all $x, y \in X$:

$$x \succeq y \iff u(x) \geq u(y).$$

Utility is **ordinal**: any strictly increasing transformation $v = f(u)$ with $f' > 0$ represents the same preferences.

Remark.

Common utility functions for two goods (x_1, x_2) :

Type	$u(x_1, x_2)$	MRS	Substitutability
Cobb-Douglas	$x_1^\alpha x_2^\beta$	$\frac{\alpha x_2}{\beta x_1}$	Diminishing
Perfect substitutes	$ax_1 + bx_2$	a/b (constant)	Perfect
Perfect complements	$\min\{ax_1, bx_2\}$	0 or ∞	Zero
Quasilinear	$v(x_1) + x_2$	$v'(x_1)$	No income effect on x_1
CES	$(x_1^\rho + x_2^\rho)^{1/\rho}$	$(x_2/x_1)^{1-\rho}$	$\sigma = 1/(1-\rho)$

Definition 1.1.3: Marginal Rate of Substitution

The MRS at a bundle (x_1, x_2) is the rate at which the consumer is willing to trade x_2 for x_1 along an indifference curve:

$$\text{MRS}_{12} = -\left.\frac{dx_2}{dx_1}\right|_{u=\bar{u}} = \frac{\partial u/\partial x_1}{\partial u/\partial x_2} = \frac{MU_1}{MU_2}.$$

Convex preferences imply a **diminishing** MRS.

1.2 Budget Constraint and Consumer Optimum**Definition 1.2.1: Budget Constraint**

With prices (p_1, p_2) and income m , the budget set is

$$\mathcal{B}(p, m) = \{(x_1, x_2) \in \mathbb{R}_+^2 : p_1x_1 + p_2x_2 \leq m\}.$$

The budget line $p_1x_1 + p_2x_2 = m$ has slope $-p_1/p_2$ and intercepts m/p_1 and m/p_2 .

Theorem 1.2.2: Consumer Optimum (Interior Solution)

If preferences are strictly convex and differentiable, the optimal bundle (x_1^*, x_2^*) satisfies the budget constraint with equality and

$$\text{MRS}_{12} = \frac{p_1}{p_2} \quad \Longleftrightarrow \quad \frac{MU_1}{MU_2} = \frac{p_1}{p_2} \quad \Longleftrightarrow \quad \frac{MU_1}{p_1} = \frac{MU_2}{p_2}.$$

The last form says that at the optimum, the marginal utility per dollar spent is equalised across all goods (**equal bang per buck**).

Proof

The Lagrangian is $\mathcal{L} = u(x_1, x_2) - \lambda(p_1x_1 + p_2x_2 - m)$. First-order conditions: $\partial u/\partial x_i = \lambda p_i$ for $i = 1, 2$. Dividing: $(\partial u/\partial x_1)/(\partial u/\partial x_2) = p_1/p_2$. $\square$

1.3 Demand Functions**Definition 1.3.1: Marshallian (Ordinary) Demand**

The solution to $\max_x u(x)$ s.t. $p \cdot x = m$ gives the Marshallian demand functions $x_i^*(p_1, p_2, m)$. For Cobb-Douglas $u = x_1^\alpha x_2^\beta$:

$$x_1^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{m}{p_1}, \quad x_2^* = \frac{\beta}{\alpha + \beta} \cdot \frac{m}{p_2}.$$

The expenditure share on good i is constant: $p_i x_i^*/m = \alpha_i/(\alpha + \beta)$.

Definition 1.3.2: Hicksian (Compensated) Demand

The solution to $\min_x p \cdot x$ s.t. $u(x) \geq \bar{u}$ gives the Hicksian demand $h_i(p, \bar{u})$. By **Shephard's Lemma**:

$$h_i(p, \bar{u}) = \frac{\partial e(p, \bar{u})}{\partial p_i},$$

where $e(p, \bar{u})$ is the **expenditure function** (minimum cost to achieve utility $\bar{u}$ at prices p).

Theorem 1.3.3: Slutsky Equation

The total price effect decomposes into a substitution effect and an income effect:

$$\underbrace{\frac{\partial x_i^*}{\partial p_j}}_{\text{total}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\substack{\text{substitution} \\ \text{(always } \leq 0 \text{ if } i=j)}} - \underbrace{x_j^* \frac{\partial x_i^*}{\partial m}}_{\text{income}}.$$

For a **normal good**, both effects reinforce (demand falls as p_i rises). For a **Giffen good**, the income effect dominates and demand rises with price.

Remark.

The Slutsky matrix $S_{ij} = \partial h_i / \partial p_j$ is symmetric and negative semi-definite. Own-price substitution effects are always non-positive: $S_{ii} \leq 0$.

1.4 Revealed Preference and Welfare**Definition 1.4.1: Indirect Utility and Roy's Identity**

The **indirect utility function** $v(p, m) = u(x^*(p, m))$ gives maximum utility as a function of prices and income. **Roy's Identity** recovers Marshallian demand from v :

$$x_i^*(p, m) = - \frac{\partial v / \partial p_i}{\partial v / \partial m}.$$

Definition 1.4.2: Equivalent and Compensating Variation

For a price change from p^0 to p^1 :

$$CV = e(p^1, u^0) - e(p^0, u^0) = m - e(p^1, u^0),$$

$$EV = e(p^1, u^1) - e(p^0, u^1) = e(p^0, u^1) - m.$$

Both measure welfare change in monetary terms; they coincide for quasilinear preferences (where consumer surplus is an exact welfare measure).

Chapter 2

Producer Theory

2.1 Production Technology

Definition 2.1.1: Production Function

$f(K, L)$ maps capital K and labour L to maximum output q . The **marginal product** of input i is $MP_i = \partial f / \partial x_i > 0$. **Diminishing marginal product**: $\partial^2 f / \partial x_i^2 < 0$.

Definition 2.1.2: Returns to Scale

For scalar $\lambda > 1$:

- **CRS** (Constant): $f(\lambda K, \lambda L) = \lambda f(K, L)$.
- **IRS** (Increasing): $f(\lambda K, \lambda L) > \lambda f(K, L)$.
- **DRS** (Decreasing): $f(\lambda K, \lambda L) < \lambda f(K, L)$.

Cobb-Douglas $f = K^\alpha L^\beta$: CRS iff $\alpha + \beta = 1$.

Definition 2.1.3: Marginal Rate of Technical Substitution

The MRTS is the slope of an isoquant:

$$\text{MRTS}_{KL} = - \frac{dK}{dL} \Big|_{q=\bar{q}} = \frac{MP_L}{MP_K}.$$

Diminishing MRTS corresponds to convex isoquants (standard assumption).

2.2 Cost Minimisation

Theorem 2.2.1: Cost-Minimising Input Choice

The firm solves $\min_{K,L} wL + rK$ s.t. $f(K, L) = q$. Interior solution satisfies:

$$\frac{MP_L}{w} = \frac{MP_K}{r} \iff \text{MRTS}_{KL} = \frac{w}{r}.$$

Definition 2.2.2: Cost Function

The **total cost function** $c(q, w, r)$ is the minimum expenditure to produce q units. By **Shephard's Lemma**:

$$\frac{\partial c}{\partial w} = L^*(q, w, r), \quad \frac{\partial c}{\partial r} = K^*(q, w, r).$$

Cost components:

$$TC = FC + VC(q), \quad AC(q) = \frac{TC}{q}, \quad MC(q) = \frac{dTC}{dq}.$$

The MC curve intersects the AC curve at the minimum of AC .

Remark.

For Cobb-Douglas $f = K^\alpha L^\beta$ with $\alpha + \beta = 1$: $c(q, w, r) = q \cdot \text{const} \cdot w^\beta r^\alpha$, so $MC = AC = \text{constant}$ (CRS $\Rightarrow$ flat LRAC).

2.3 Profit Maximisation

Definition 2.3.1: Profit Function

$\pi = TR - TC = pq - c(q)$. Under perfect competition, p is taken as given. FOC: $p = MC(q^*)$. SOC: $MC'(q^*) > 0$ (rising MC). The firm shuts down if $p < \min AVC$ (short run) or $p < \min AC$ (long run).

Theorem 2.3.2: Hotelling's Lemma

If $\pi(p, w, r) = \max_{q, K, L} [pf(K, L) - wL - rK]$, then:

$$\frac{\partial \pi}{\partial p} = q^*, \quad \frac{\partial \pi}{\partial w} = -L^*, \quad \frac{\partial \pi}{\partial r} = -K^*.$$

Chapter 3

Market Structures

3.1 Perfect Competition

Definition 3.1.1: Competitive Equilibrium

Price p^* and quantity Q^* such that $Q^D(p^*) = Q^S(p^*)$. Each firm takes p^* as given and produces where $p^* = MC(q_i)$. In long-run equilibrium: $p^* = MC = \min AC$ (zero economic profit).

Definition 3.1.2: Consumer and Producer Surplus

$$CS = \int_0^{Q^*} [P^D(Q) - p^*] dQ, \quad PS = \int_0^{Q^*} [p^* - P^S(Q)] dQ.$$

Total Surplus = $CS + PS$ is maximised at competitive equilibrium (**First Welfare Theorem**).

3.2 Monopoly

Definition 3.2.1: Monopoly Optimum

The monopolist maximises $\pi = TR - TC = p(Q) \cdot Q - c(Q)$. FOC: $MR(Q^*) = MC(Q^*)$, where $MR = p + p'(Q)Q = p(1 - 1/|\varepsilon|)$. The **Lerner Index** measures market power:

$$L = \frac{p - MC}{p} = \frac{1}{|\varepsilon_D|}.$$

Deadweight loss (DWL) equals the welfare lost relative to the competitive outcome.

Remark.

Price discrimination: (1st degree) personalized prices extracting all CS; (2nd degree) nonlinear pricing/quantity discounts; (3rd degree) separate markets segmented by ε . Under 3rd degree: $MR_1 = MR_2 = MC$.

3.3 Oligopoly

Definition 3.3.1: Cournot Competition

n firms simultaneously choose quantities q_i . Firm i maximises:

$$\pi_i = p(Q)q_i - c_i(q_i), \quad Q = \sum_j q_j.$$

FOC (best response): $p(Q) + p'(Q)q_i = MC_i$. Symmetric equilibrium with linear demand $p = a - bQ$ and constant MC :

$$q^* = \frac{a - MC}{b(n + 1)}, \quad p^* = \frac{a + n \cdot MC}{n + 1}.$$

As $n \rightarrow \infty$: $p^* \rightarrow MC$ (competitive outcome).

Definition 3.3.2: Bertrand Competition

Firms simultaneously set prices. With homogeneous goods and constant identical MC : unique Nash equilibrium is $p_1 = p_2 = MC$ (Bertrand paradox). With differentiated products or capacity constraints, equilibrium prices exceed MC .

Definition 3.3.3: Stackelberg Competition

Leader (firm 1) commits to quantity q_1 first; follower (firm 2) best-responds. Backward induction: leader internalises the follower's reaction function $q_2^{BR}(q_1)$. Leader has first-mover advantage: higher profit than Cournot.

Chapter 4

Welfare Economics and Market Failures

4.1 Welfare Analysis

Theorem 4.1.1: First Welfare Theorem

If all markets are perfectly competitive and there are no externalities or public goods, then every competitive equilibrium is **Pareto efficient**: no reallocation can make someone better off without making someone else worse off.

Theorem 4.1.2: Second Welfare Theorem

Under convexity assumptions, every Pareto-efficient allocation can be supported as a competitive equilibrium with appropriate lump-sum transfers of endowments.

4.2 Externalities and Public Goods

Definition 4.2.1: Externality

A cost or benefit imposed on a third party not involved in the transaction.

- **Negative externality**: $MSC > MPC$ (e.g. pollution). Optimal Pigouvian tax $t^* = MSC - MPC$ at efficient output.
- **Positive externality**: $MSB > MPB$ (e.g. vaccination). Optimal Pigouvian subsidy $s^* = MSB - MPB$.

Theorem 4.2.2: Coase Theorem

If property rights are well-defined and bargaining is costless, private negotiation leads to the efficient outcome regardless of the initial assignment of rights. In practice, transaction costs, incomplete information, and hold-up problems limit its applicability.

Definition 4.2.3: Public Good

A good that is **non-excludable** (cannot prevent consumption) and **non-rival** (one person's consumption does not reduce another's). Samuelson rule for efficient provision: $\sum_i MRS_i = MRT$ (sum of marginal willingness to pay equals marginal cost of provision). Market under-provides due to the **free-rider problem**.

Chapter 5

Information Economics

5.1 Adverse Selection

Definition 5.1.1: Adverse Selection

Arises when one party has **hidden information** (pre-contractual), causing a market to attract a disproportionately unfavourable pool. Classic example (Akerlof): in the used-car market with quality $\theta \sim U[0, 1]$, sellers know θ , buyers do not. At any pooling price p , only sellers with $\theta \leq p/v_s$ sell; buyers anticipate average quality $p/(2v_s)$, driving the market toward collapse.

Definition 5.1.2: Signalling Equilibrium

The high-type agent undertakes a costly signal s to credibly distinguish itself. Spence education model: worker of type $\theta \in \{H, L\}$ chooses education e . Separating equilibrium requires the single-crossing (Spence-Mirrlees) condition: $\partial^2 u / (\partial e \partial \theta) > 0$, ensuring the incentive-compatibility constraints are not both binding simultaneously.

$$IC_H : u(w_H, e_H; H) \geq u(w_L, e_L; H), \quad IC_L : u(w_L, e_L; L) \geq u(w_H, e_H; L).$$

5.2 Moral Hazard

Definition 5.2.1: Moral Hazard

Arises when one party's **hidden action** (post-contractual) cannot be observed or verified. Principal-agent problem: principal maximises $E[q(e)] - w(q)$ subject to the agent's

$$IR : E[u(w(q))] - c(e) \geq \bar{u} \quad (\text{participation}),$$

$$IC : e^* = \arg \max_e E[u(w(q))] - c(e) \quad (\text{incentive compatibility}).$$

First-best requires both risk sharing and incentive provision; with risk-averse agents these goals conflict (**second-best** involves incomplete insurance).

Remark.

Key results: (1) If agent is risk-neutral, sell the firm (franchise contract) achieves first-best. (2) Optimal second-best contract balances insurance against incentives; bonus/commission schemes approximate it. (3) Relative performance evaluation (RPE) removes common shocks from incentive pay.

Chapter 6

Choice Under Uncertainty

6.1 Expected Utility Theory

Definition 6.1.1: Von Neumann–Morgenstern Expected Utility

A lottery $L = (p_1, x_1; \dots; p_n, x_n)$ assigns probability p_i to outcome x_i . Under the VNM axioms (completeness, transitivity, continuity, independence), there exists a Bernoulli utility function $u : \mathbb{R} \rightarrow \mathbb{R}$ such that:

$$U(L) = \sum_{i=1}^n p_i u(x_i) = E[u(x)].$$

u is unique up to positive affine transformations.

Definition 6.1.2: Risk Aversion

An agent is **risk-averse** if u is strictly concave: $u'' < 0$.

- **Certainty equivalent CE :** $u(CE) = E[u(x)]$, so $CE \leq E[x]$.
- **Risk premium $\pi_R = E[x] - CE \geq 0$.**
- **Arrow-Pratt absolute risk aversion:** $A(x) = -u''(x)/u'(x)$.
- **Relative risk aversion:** $R(x) = -xu''(x)/u'(x)$.

For CARA: $u(x) = -e^{-\alpha x}$, so $A(x) = \alpha$ (constant). For CRRA: $u(x) = x^{1-\rho}/(1-\rho)$, so $R(x) = \rho$ (constant).

Theorem 6.1.3: Jensen's Inequality

For a concave function u and random variable x : $u(E[x]) \geq E[u(x)]$. Equality holds iff x is degenerate (no risk).

Part II

Macroeconomics

Chapter 7

National Income and Output

7.1 GDP and National Accounts

Definition 7.1.1: Gross Domestic Product

GDP measures the total market value of all **final goods and services** produced within a country's borders in a given period. Three equivalent approaches:

$$GDP = C + I + G + NX \quad (\text{expenditure approach}),$$

where C = private consumption, I = gross investment, G = government spending, $NX = X - M$ = net exports.

Definition 7.1.2: Real vs. Nominal GDP

Nominal GDP values output at current prices. **Real GDP** values output at base-year prices, removing the effect of price changes.

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100.$$

Growth in real GDP $\approx$ growth in nominal GDP $-$ inflation.

Remark.

GDP captures production but misses distribution, non-market activity, informal economy, and welfare. GNP adds net factor income from abroad.

7.2 Inflation and Price Levels

Definition 7.2.1: CPI and Inflation

The Consumer Price Index tracks the cost of a fixed basket of goods:

$$\text{CPI}_t = \frac{\sum p_t^i q_0^i}{\sum p_0^i q_0^i} \times 100.$$

Inflation rate: $\pi_t = (\text{CPI}_t - \text{CPI}_{t-1})/\text{CPI}_{t-1}$. **Fisher equation** linking nominal rate i , real rate r , and inflation π :

$$(1 + i) = (1 + r)(1 + \pi) \approx r + \pi \quad (\text{for small rates}).$$

Chapter 8

Money and Monetary Policy

8.1 Money Supply and Demand

Definition 8.1.1: Money and its Functions

Money serves as (1) medium of exchange, (2) unit of account, (3) store of value. **M1** = currency + demand deposits. **M2** = M1 + savings deposits + small time deposits. **Money multiplier**: $m = 1/\text{reserve ratio}$ (simple model); in practice, $m = (1 + c)/(c + \theta)$ where c = currency-to-deposit ratio, θ = reserve ratio.

8.2 Monetary Policy and the Interest Rate

Remark.

Central banks (e.g., the Fed) target the short-term nominal interest rate (federal funds rate) using open market operations (buying/selling government bonds). Expansionary policy: buy bonds $\Rightarrow$ money supply $\uparrow \Rightarrow i \downarrow \Rightarrow$ investment $\uparrow \Rightarrow$ GDP $\uparrow$. Key constraint: zero lower bound (ZLB); unconventional tools include QE (asset purchases) and forward guidance.

Chapter 9

Fiscal Policy

9.1 Government Budget and Multiplier Effects

Definition 9.1.1: Fiscal Multiplier

An increase in government spending ΔG raises GDP by more than ΔG :

$$\Delta Y = \frac{1}{1 - MPC} \cdot \Delta G = \frac{1}{MPS} \cdot \Delta G,$$

where MPC is the marginal propensity to consume and $MPS = 1 - MPC$. Tax multiplier: $\Delta Y = -\frac{MPC}{1 - MPC} \cdot \Delta T$ (smaller in magnitude; negative sign because tax cut raises disposable income indirectly). **Balanced budget multiplier** = 1 (equal ΔG and ΔT).

Remark.

Crowding out: expansionary fiscal policy may raise interest rates, reducing private investment. In a liquidity trap (i at ZLB), crowding out is absent and the multiplier is larger. Long-run: deficit spending accumulates debt (D/GDP ratio); sustainability requires $r < g$ (interest rate below growth rate).

Chapter 10

International Economics

10.1 Exchange Rates and the Balance of Payments

Definition 10.1.1: Nominal and Real Exchange Rate

The **nominal exchange rate** e is units of domestic currency per unit of foreign currency. The **real exchange rate**:

$$q = e \cdot \frac{P^*}{P},$$

where P^* is the foreign price level and P the domestic price level. A rise in q is a **real depreciation** (domestic goods become cheaper). **Purchasing Power Parity (PPP)**: $q = 1$ in the long run, implying $e = P/P^*$.

Definition 10.1.2: Balance of Payments

$$\underbrace{CA}_{\text{Current Account}} + \underbrace{FA}_{\text{Financial Account}} = 0.$$

$CA = NX + \text{net factor income} + \text{net transfers}$. A current account **deficit** means the country imports more than it exports and is a net borrower from the rest of the world.

Remark.

Under fixed exchange rates, the central bank must intervene (buy/sell FX reserves) to maintain the peg; monetary policy autonomy is lost (**impossible trinity**: cannot simultaneously have fixed rate, free capital flows, and independent monetary policy).

Part III

Financial Accounting

Chapter 11

Reading Financial Statements

11.1 The Three Core Statements

Definition 11.1.1: Income Statement

Reports **revenues** and **expenses** over a period, resulting in net income (profit or loss):

$$\text{Net Income} = \text{Revenue} - \text{COGS} - \text{Operating Expenses} - \text{Interest} - \text{Taxes}.$$

Key line items: Revenue, Cost of Goods Sold (COGS), **Gross Profit** = Revenue – COGS, **EBITDA**, **EBIT** (Operating Income), **EBT**, **Net Income**.

Definition 11.1.2: Balance Sheet (Statement of Financial Position)

A snapshot of what the firm **owns** (assets), **owes** (liabilities), and the residual claim of shareholders (equity) at a point in time:

$$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}.$$

Current assets (cash, AR, inventory) vs. **non-current assets** (PP&E, intangibles, goodwill). **Current liabilities** (AP, short-term debt) vs. **long-term debt**.

Definition 11.1.3: Statement of Cash Flows

Reconciles net income to actual cash generated:

$$\text{Cash Flow} = \underbrace{CFO}_{\text{Operations}} + \underbrace{CFI}_{\text{Investing}} + \underbrace{CFF}_{\text{Financing}}.$$

CFO: net income + D&A – Δ Working Capital (indirect method). **CFI**: capex, acquisitions, asset disposals. **CFF**: debt issuance/repayment, equity issuance, dividends. **Free Cash Flow**: $FCF = CFO - \text{Capex}$.

11.2 Key Ratios for Analysts

Ratio	Formula	Measures
<i>Profitability</i>		
Gross Margin	Gross Profit / Revenue	Pricing power
EBITDA Margin	EBITDA / Revenue	Operating efficiency
ROE	Net Income / Avg. Equity	Return to shareholders
ROA	Net Income / Avg. Assets	Asset efficiency
<i>Leverage</i>		
Debt/Equity	Total Debt / Total Equity	Capital structure
Interest Coverage	EBIT / Interest Expense	Ability to service debt
<i>Liquidity</i>		
Current Ratio	Current Assets / Current Liabilities	Short-term solvency
Quick Ratio	(CA – Inventory) / CL	Stringent liquidity
<i>Efficiency</i>		
Asset Turnover	Revenue / Avg. Assets	Asset productivity
Inventory Days	(Inventory / COGS) × 365	Inventory management

Remark.

DuPont decomposition:
$$ROE = \underbrace{\frac{NI}{Sales}}_{\text{margin}} \times \underbrace{\frac{Sales}{Assets}}_{\text{turnover}} \times \underbrace{\frac{Assets}{Equity}}_{\text{leverage}}.$$

11.3 Accrual Accounting vs. Cash Accounting

Definition 11.3.1: Accrual Basis

Revenue is recognised when **earned** (goods delivered/services rendered), not when cash is received. Expenses are recognised when **incurred**, not when paid. Result: net income $\neq$ cash flow in any given period. Key accruals:

- **Accounts Receivable:** revenue earned, cash not yet received.
- **Accounts Payable:** expense incurred, cash not yet paid.
- **Deferred Revenue:** cash received, revenue not yet earned.
- **Prepaid Expense:** cash paid, expense not yet incurred.

Part IV

Corporate Finance

Chapter 12

Time Value of Money

12.1 Discounting and Compounding

Definition 12.1.1: Present and Future Value

A cash flow C received in n periods, discounted at rate r per period:

$$PV = \frac{C}{(1+r)^n}, \quad FV = C \cdot (1+r)^n.$$

Continuous compounding: $PV = C e^{-rn}$, $FV = C e^{rn}$.

Definition 12.1.2: Annuities and Perpetuities

Ordinary annuity (payments at end of period):

$$PV_{\text{annuity}} = PMT \cdot \frac{1 - (1+r)^{-n}}{r}.$$

Growing annuity (payment grows at rate g , $g \neq r$):

$$PV = PMT \cdot \frac{1 - [(1+g)/(1+r)]^n}{r - g}.$$

Perpetuity: $PV = PMT/r$. **Growing perpetuity** (Gordon Growth Model): $PV = PMT/(r - g)$, $g < r$.

Definition 12.1.3: Effective Annual Rate

For m compounding periods per year at nominal rate r_{nom} :

$$EAR = \left(1 + \frac{r_{\text{nom}}}{m}\right)^m - 1.$$

Continuous: $EAR = e^{r_{\text{nom}}} - 1$.

Chapter 13

Capital Budgeting

13.1 Investment Decision Rules

Definition 13.1.1: Net Present Value

$$NPV = \sum_{t=0}^T \frac{CF_t}{(1+r)^t} = -C_0 + \sum_{t=1}^T \frac{CF_t}{(1+r)^t}.$$

NPV Rule: accept project iff $NPV > 0$. NPV correctly accounts for the time value of money, risk, and opportunity cost of capital.

Definition 13.1.2: Internal Rate of Return

The IRR solves $NPV(IRR) = 0$:

$$0 = \sum_{t=0}^T \frac{CF_t}{(1+IRR)^t}.$$

IRR Rule: accept iff $IRR > \text{hurdle rate (cost of capital)}$. Pitfalls: (1) multiple IRRs if sign changes more than once; (2) scale problem (higher IRR $\nRightarrow$ higher NPV); (3) assumes reinvestment at IRR. Always prefer NPV for mutually exclusive projects.

Remark.

Other metrics: **Payback Period** (time to recoup C_0) ignores TVM and post-payback flows. **Profitability Index** $= NPV/C_0$ useful for capital rationing (ranking projects under budget constraint).

Chapter 14

Risk and Cost of Capital

14.1 Weighted Average Cost of Capital

Definition 14.1.1: WACC

The after-tax required return on the firm's assets, weighted by capital structure:

$$WACC = \frac{E}{V}r_E + \frac{D}{V}r_D(1 - \tau_C),$$

where $V = E + D$ is total firm value, r_E is cost of equity, r_D is pre-tax cost of debt, and τ_C is the corporate tax rate. Use WACC to discount FCFs when capital structure is stable.

Definition 14.1.2: Cost of Equity via CAPM

$$r_E = r_f + \beta_E \cdot [E(R_M) - r_f].$$

β_E (equity beta) incorporates both business risk and financial leverage. **Unlevering:** $\beta_A = \frac{E}{E+D(1-\tau_C)}\beta_E + \frac{D(1-\tau_C)}{E+D(1-\tau_C)}\beta_D \approx \frac{E}{E+D}\beta_E$ (if $\beta_D \approx 0$).

Chapter 15

Capital Structure

15.1 Modigliani–Miller Theorems

Theorem 15.1.1: MM Proposition I (No Taxes)

In perfect capital markets (no taxes, no bankruptcy costs, no information asymmetry, no transaction costs), the value of a firm is independent of its capital structure:

$$V_L = V_U.$$

Theorem 15.1.2: MM Proposition II (No Taxes)

The cost of equity is an increasing linear function of leverage:

$$r_E = r_A + \frac{D}{E}(r_A - r_D),$$

where r_A is the unlevered cost of equity. WACC is constant regardless of leverage.

Theorem 15.1.3: MM with Corporate Taxes

Interest payments are tax-deductible, creating a **tax shield**:

$$V_L = V_U + \tau_C D \quad (\text{assuming perpetual debt}).$$

Implication: firms should use maximum leverage. Real-world limits: financial distress costs and agency costs.

Definition 15.1.4: Trade-Off Theory

Optimal capital structure balances the tax shield benefit of debt against financial distress costs:

$$V_L = V_U + PV(\text{Tax Shield}) - PV(\text{Distress Costs}).$$

Predicts an interior optimal leverage ratio for each firm.

Definition 15.1.5: Pecking Order Theory

Firms prefer internal financing (retained earnings), then debt, then equity. Rationale: adverse selection costs are lowest for internal funds, moderate for debt, and highest for equity (Myers-Majluf).

Part V

Investments and Portfolio Theory

Chapter 16

Financial Markets and Return Measurement

16.1 Asset Classes and Market Structure

Definition 16.1.1: Primary vs. Secondary Markets

Primary market: new securities are issued (IPO, seasoned offering). **Secondary market:** previously issued securities trade between investors (NYSE, NASDAQ). Market efficiency implies secondary prices reflect all available information.

16.2 Return Measurement

Definition 16.2.1: Holding Period Return

For a single period with beginning price P_0 , ending price P_1 , and dividend D_1 :

$$HPR = \frac{P_1 - P_0 + D_1}{P_0} = \frac{P_1 + D_1}{P_0} - 1.$$

Definition 16.2.2: Multi-Period Returns

Arithmetic mean (best estimate of expected single-period return):

$$\bar{R}_A = \frac{1}{T} \sum_{t=1}^T R_t.$$

Geometric mean (best estimate of compound growth rate):

$$\bar{R}_G = \left(\prod_{t=1}^T (1 + R_t) \right)^{1/T} - 1.$$

Always $\bar{R}_G \leq \bar{R}_A$; equality iff all returns are equal. Approximation: $\bar{R}_G \approx \bar{R}_A - \sigma^2/2$.

Definition 16.2.3: Risk Measures

Sample variance and standard deviation of returns:

$$s^2 = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2, \quad s = \sqrt{s^2}.$$

Sharpe Ratio: $SR = (E[R] - r_f)/\sigma$ (reward per unit of total risk).

Chapter 17

Portfolio Theory

17.1 Mean-Variance Analysis

Definition 17.1.1: Portfolio Return and Variance

For n assets with weights $\mathbf{w} = (w_1, \dots, w_n)^\top$, $\mathbf{1}^\top \mathbf{w} = 1$:

$$E[R_p] = \mathbf{w}^\top \boldsymbol{\mu} = \sum_{i=1}^n w_i \mu_i,$$

$$\sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij},$$

where $\boldsymbol{\Sigma}$ is the $n \times n$ covariance matrix and $\sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$.

Definition 17.1.2: Diversification

For an equally weighted portfolio of n assets with average variance $\bar{\sigma}^2$ and average covariance $\bar{\sigma}_{ij}$:

$$\sigma_p^2 = \frac{1}{n} \bar{\sigma}^2 + \frac{n-1}{n} \bar{\sigma}_{ij} \xrightarrow{n \rightarrow \infty} \bar{\sigma}_{ij}.$$

Idiosyncratic (diversifiable) risk vanishes; systematic risk remains.

17.2 Efficient Frontier

Definition 17.2.1: Minimum Variance Portfolio

The **Global Minimum Variance (GMV)** portfolio solves:

$$\min_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1.$$

Solution: $\mathbf{w}_{GMV} = \frac{\boldsymbol{\Sigma}^{-1} \mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}}$. Adding the return target $\mathbf{w}^\top \boldsymbol{\mu} = \mu_p$ traces the **efficient frontier** in $(\sigma, E[R])$ space.

Definition 17.2.2: Capital Allocation Line and Tangency Portfolio

With a risk-free asset at rate r_f , the investor's opportunity set becomes the **Capital Allocation Line (CAL)**:

$$E[R_p] = r_f + \frac{E[R_T] - r_f}{\sigma_T} \sigma_p,$$

where T is the **tangency portfolio** maximising the Sharpe ratio:

$$\mathbf{w}_T = \frac{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f \mathbf{1})}.$$

Theorem 17.2.3: Two-Fund Separation Theorem

Every mean-variance efficient portfolio is a linear combination of the risk-free asset and the tangency portfolio. All investors, regardless of risk aversion, choose a portfolio on the CAL through T .

Definition 17.2.4: Investor Utility and Optimal Allocation

For mean-variance utility $U = E[R_p] - \frac{A}{2} \sigma_p^2$, the optimal fraction w^* invested in the risky asset (or risky portfolio) satisfies:

$$w^* = \frac{E[R] - r_f}{A \sigma^2}.$$

Higher risk aversion (A) or higher variance $\Rightarrow$ lower allocation to risky assets.

Chapter 18

CAPM and Factor Models

18.1 Capital Asset Pricing Model

Definition 18.1.1: CAPM

In equilibrium, all investors hold the market portfolio M . For any asset i :

$$E[R_i] = r_f + \beta_i (E[R_M] - r_f),$$

where the **beta** measures systematic risk:

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)} = \frac{\sigma_{iM}}{\sigma_M^2}.$$

The **Security Market Line (SML)** plots $E[R_i]$ against β_i . The **Capital Market Line (CML)** plots $E[R_p]$ against σ_p for efficient (fully diversified) portfolios:

$$E[R_p] = r_f + \frac{E[R_M] - r_f}{\sigma_M} \sigma_p.$$

Definition 18.1.2: Risk Decomposition and Alpha

Total variance decomposes into systematic and idiosyncratic parts:

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon_i}^2.$$

Alpha (Jensen's alpha) measures risk-adjusted outperformance:

$$\alpha_i = E[R_i] - [r_f + \beta_i (E[R_M] - r_f)].$$

In an efficient market, $\alpha = 0$ for all assets.

18.2 Factor Models

Definition 18.2.1: Fama-French Three-Factor Model

Extends CAPM with size and value factors:

$$E[R_i] - r_f = \alpha_i + \beta_{iM}(R_M - r_f) + \beta_{iSMB} \cdot SMB + \beta_{iHML} \cdot HML,$$

where SMB (Small Minus Big) captures the size premium, and HML (High Minus Low book-to-market) captures the value premium. Carhart adds MOM (momentum).

Remark.

The CAPM assumptions that fail empirically: (1) all investors hold the same mean-variance efficient portfolio; (2) no taxes or transaction costs; (3) unlimited borrowing at r_f . Factor models are more descriptive; they do not necessarily imply risk-based explanations (may reflect mispricing or data mining).

Chapter 19

Equity Valuation

19.1 Dividend Discount Model

Definition 19.1.1: DDM (Gordon Growth Model)

The price of a stock equals the PV of all future dividends:

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + r_E)^t}.$$

For constant growth g (Gordon Growth Model):

$$P_0 = \frac{D_1}{r_E - g} = \frac{D_0(1 + g)}{r_E - g}, \quad g < r_E.$$

Implies: $r_E = D_1/P_0 + g$ (dividend yield + capital gains yield).

19.2 Free Cash Flow to Firm (FCFF) Model

Definition 19.2.1: DCF Valuation

Enterprise value equals the PV of future FCFs discounted at WACC:

$$EV = \sum_{t=1}^T \frac{FCFF_t}{(1 + WACC)^t} + \frac{TV_T}{(1 + WACC)^T},$$

where terminal value $TV_T = FCFF_{T+1}/(WACC - g)$ (Gordon Growth).

$$\text{Equity Value} = EV - \text{Net Debt}.$$

$$FCFF = EBIT(1 - \tau) + D\&A - \Delta NWC - \text{Capex}.$$

19.3 Relative Valuation

Multiple	Formula	Use
P/E	Price per share / EPS	Mature, profitable firms
EV/EBITDA	Enterprise Value / EBITDA	Cross-capital-structure comparisons
EV/Sales	Enterprise Value / Revenue	High-growth or loss-making firms
P/B	Price / Book Value per share	Financials; asset-heavy sectors
PEG	P/E / Earnings Growth Rate	Growth-adjusted P/E

Remark.

Multiples are only meaningful relative to peers. Always adjust for differences in growth, risk, and margins across comparables.

Chapter 20

Fixed Income

20.1 Bond Pricing

Definition 20.1.1: Bond Price

A bond with face value F , coupon rate c , maturity T , and yield to maturity y has price:

$$P = \sum_{t=1}^T \frac{cF}{(1+y)^t} + \frac{F}{(1+y)^T} = cF \cdot \frac{1 - (1+y)^{-T}}{y} + F(1+y)^{-T}.$$

$P > F$ iff $y < c$ (premium bond); $P < F$ iff $y > c$ (discount bond). Price and yield move **inversely**.

20.2 Duration and Convexity

Definition 20.2.1: Macaulay Duration

The weighted-average time to receipt of cash flows (weights = PV of each cash flow as fraction of total price):

$$D_{Mac} = \frac{1}{P} \sum_{t=1}^T t \cdot \frac{CF_t}{(1+y)^t}.$$

Modified Duration: $D_{Mod} = D_{Mac}/(1+y)$. Price sensitivity: $\Delta P \approx -D_{Mod} \cdot P \cdot \Delta y$.

Definition 20.2.2: Convexity

Second-order correction to the price-yield relationship:

$$\frac{\Delta P}{P} \approx -D_{Mod} \Delta y + \frac{1}{2} C (\Delta y)^2,$$

where convexity $C = \frac{1}{P} \frac{d^2 P}{dy^2}$. Higher convexity is always beneficial: the bond rises more (falls less) than the duration approximation predicts for any yield change.

20.3 Yield Curve

Definition 20.3.1: Term Structure of Interest Rates

The **yield curve** plots spot rates s_t (YTM of zero-coupon bonds) against maturity t . Shapes: normal (upward sloping), flat, inverted. **Forward rate** $f_{t,t+1}$ is the implied one-period rate from t to $t+1$:

$$(1 + s_{t+1})^{t+1} = (1 + s_t)^t (1 + f_{t,t+1}).$$

Theories: (1) Expectations: $f_{t,t+1} = E[s_{t+1}]$; (2) Liquidity preference: $f_{t,t+1} > E[s_{t+1}]$ (term premium); (3) Market segmentation: supply/demand at each maturity.

Chapter 21

Derivatives and Options

21.1 Option Payoffs and Strategies

Definition 21.1.1: Call and Put Options

A **European call** gives the right to buy asset S at strike K at maturity T . Payoff at expiry: $\max(S_T - K, 0)$.
A **European put**: payoff $\max(K - S_T, 0)$. Premium (price) paid upfront; net profit = payoff – premium.

Definition 21.1.2: Put–Call Parity

For European options on a non-dividend-paying stock:

$$C - P = S_0 - Ke^{-rT},$$

where C and P are call and put prices with the same K and T . Violation implies arbitrage.

Strategy	Construction	View
Covered Call	Long stock + Short call	Mildly bullish, income
Protective Put	Long stock + Long put	Bullish, insured
Bull Spread	Long call(K_1) + Short call(K_2), $K_1 < K_2$	Moderately bullish
Bear Spread	Long put(K_2) + Short put(K_1)	Moderately bearish
Straddle	Long call + Long put (same K, T)	High volatility
Strangle	Long call(K_H) + Long put(K_L), $K_H > K_L$	High vol, cheaper

21.2 Black-Scholes-Merton Model

Definition 21.2.1: BSM Price

Assuming the stock price follows geometric Brownian motion $dS = \mu S dt + \sigma S dW_t$, the no-arbitrage price of a European call is:

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2),$$
$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

where $N(\cdot)$ is the standard normal CDF. Put price: $P = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$.

Definition 21.2.2: The Greeks

Greek	Definition	Interpretation
Delta Δ	$\partial C / \partial S$	Change in option price per \$1 change in S ; $\in (0, 1)$ for calls
Gamma Γ	$\partial^2 C / \partial S^2$	Rate of change of delta; highest near ATM
Theta Θ	$\partial C / \partial t$	Time decay; typically negative for long options
Vega ν	$\partial C / \partial \sigma$	Sensitivity to volatility; positive for long options
Rho ρ	$\partial C / \partial r$	Sensitivity to interest rate

Part VI

Quantitative Finance

Chapter 22

Portfolio Optimisation

22.1 Mean-Variance Optimisation

Definition 22.1.1: Markowitz Optimisation Problem

For a target return μ_p^* , the minimum variance portfolio solves:

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \boldsymbol{\mu} = \mu_p^*, \quad \mathbf{1}^\top \mathbf{w} = 1.$$

Using Lagrangians λ and γ , the KKT conditions give the closed form:

$$\mathbf{w}^* = \lambda^* \Sigma^{-1} \boldsymbol{\mu} + \gamma^* \Sigma^{-1} \mathbf{1},$$

where the multipliers solve the 2×2 system:

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} \lambda^* \\ \gamma^* \end{pmatrix} = \begin{pmatrix} \mu_p^* \\ 1 \end{pmatrix},$$

with $A = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, $B = \boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{1}$, $C = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, and $\Delta = AC - B^2 > 0$. The minimum variance of the frontier portfolio at μ_p^* is:

$$\sigma^{*2}(\mu_p^*) = \frac{C\mu_p^{*2} - 2B\mu_p^* + A}{\Delta}.$$

22.2 Sharpe Ratio Maximisation

Definition 22.2.1: Tangency Portfolio (Analytical Form)

With a risk-free rate r_f , the excess return vector is $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - r_f \mathbf{1}$. The tangency (maximum Sharpe) portfolio is:

$$\mathbf{w}_T = \frac{\Sigma^{-1} \tilde{\boldsymbol{\mu}}}{\mathbf{1}^\top \Sigma^{-1} \tilde{\boldsymbol{\mu}}}, \quad SR_T = \sqrt{\tilde{\boldsymbol{\mu}}^\top \Sigma^{-1} \tilde{\boldsymbol{\mu}}}.$$

22.3 Practical Constraints and Shrinkage

Remark.

The classical Markowitz solution is notoriously unstable: small errors in $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ lead to extreme allocations.

Practical remedies:

- **Long-only constraint:** $w_i \geq 0$ (convex QP, solved via active-set or interior-point methods).
- **Shrinkage estimators:** Ledoit-Wolf shrinks $\hat{\boldsymbol{\Sigma}}$ toward a structured target (e.g., diagonal) to reduce estimation error.
- **Black-Litterman:** blends market-implied equilibrium returns with investor views via Bayesian updating, producing better-conditioned portfolios.
- **Risk parity:** allocate equal risk contribution $w_i(\boldsymbol{\Sigma}\mathbf{w})_i/\sigma_p^2 = 1/n$ across assets.

Chapter 23

Mathematical Finance

23.1 Single-Period Model and No-Arbitrage

Definition 23.1.1: Single-Period Market Model

A market consists of:

- A **risk-free bond**: value 1 at time 0, value $1 + r$ at time 1.
- d **risky assets**: price vector $S_0 \in \mathbb{R}^d$ at time 0; random payoff $S_1(\omega)$ at time 1 in state $\omega \in \Omega$ (finite state space, $|\Omega| = k$).
- A **portfolio** $H = (H^0, H^1, \dots, H^d)$: holdings of bond and risky assets.

Portfolio value: $V_0 = H^0 + H \cdot S_0$, $V_1(\omega) = H^0(1 + r) + H \cdot S_1(\omega)$.

Definition 23.1.2: Arbitrage

A portfolio H is an **arbitrage** if:

$$V_0 \leq 0, \quad V_1(\omega) \geq 0 \quad \forall \omega, \quad \exists \omega^* : V_1(\omega^*) > 0.$$

(Get something from nothing, or nothing for free with positive probability of gain.)

Theorem 23.1.3: Fundamental Theorem of Asset Pricing (FTAP) — Single Period

The market is **arbitrage-free** if and only if there exists a **risk-neutral (equivalent martingale) measure** $\mathbb{Q}$, i.e., a probability measure with $\mathbb{Q}(\omega) > 0 \quad \forall \omega$ such that

$$S_0^i = \frac{1}{1+r} \mathbb{E}^{\mathbb{Q}}[S_1^i] \quad \forall i = 1, \dots, d.$$

The market is **complete** (every payoff is replicable) iff $\mathbb{Q}$ is unique.

23.2 Binomial Model

Definition 23.2.1: One-Period Binomial Model

Stock price S_0 moves to S_0u (up) with physical probability p , or S_0d (down) with $1 - p$, where $d < 1 + r < u$ (no-arbitrage condition). The unique risk-neutral probability:

$$q = \frac{(1 + r) - d}{u - d}.$$

Price of a derivative with payoffs (V_u, V_d) :

$$V_0 = \frac{qV_u + (1 - q)V_d}{1 + r}.$$

Replicating portfolio: $\Delta = (V_u - V_d)/(S_0u - S_0d)$ shares, bond position $B = (V_0 - \Delta S_0)/(1 + r)^0$.

Definition 23.2.2: Multi-Period Binomial Tree

After n steps, each path is a sequence of ups and downs. The T -period European option price:

$$V_0 = \frac{1}{(1 + r)^T} \sum_{k=0}^T \binom{T}{k} q^k (1 - q)^{T-k} h(S_0 u^k d^{T-k}),$$

where $h(S_T)$ is the payoff function. American options: use backward induction, taking $\max(\text{intrinsic value}, \text{continuation value})$ at each node.

Chapter 24

Stochastic Processes and Black-Scholes

24.1 Brownian Motion

Definition 24.1.1: Standard Brownian Motion

A stochastic process $\{W_t\}_{t \geq 0}$ is a **standard Brownian motion** (Wiener process) if:

1. $W_0 = 0$ a.s.
2. **Independent increments:** for $0 \leq s < t$, $W_t - W_s$ is independent of $\mathcal{F}_s$ (the natural filtration).
3. **Gaussian increments:** $W_t - W_s \sim \mathcal{N}(0, t - s)$.
4. **Continuous paths:** $t \mapsto W_t$ is continuous a.s.

Remark.

Key properties of Brownian motion:

- $E[W_t] = 0$, $\text{Var}(W_t) = t$, $\text{Cov}(W_s, W_t) = \min(s, t)$.
- Paths are continuous but nowhere differentiable (a.s.).
- **Quadratic variation:** $[W]_t = t$ (i.e., $\sum (W_{t_i} - W_{t_{i-1}})^2 \rightarrow t$).
- W_t is a martingale: $E[W_t | \mathcal{F}_s] = W_s$.

24.2 Itô Calculus

Definition 24.2.1: Itô Integral

For an adapted process $\{H_t\}$, the **Itô integral** is defined as:

$$\int_0^T H_t dW_t = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} H_{t_i} (W_{t_{i+1}} - W_{t_i}),$$

using the *left-endpoint* (non-anticipating) evaluation. This differs from the Stratonovich integral and preserves the martingale property.

Theorem 24.2.2: Itô's Lemma

Let X_t be an Itô process: $dX_t = \mu_t dt + \sigma_t dW_t$. For $f \in C^{2,1}(\mathbb{R} \times [0, T])$:

$$df(X_t, t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dW_t.$$

The extra term $\frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2} dt$ arises from the quadratic variation of W_t : $(dW_t)^2 = dt$ (Itô's table: $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt$).

24.3 Geometric Brownian Motion and the BSM Equation**Definition 24.3.1: Geometric Brownian Motion**

The standard model for stock prices:

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

By Itô's lemma applied to $f(S_t) = \ln S_t$:

$$d \ln S_t = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t,$$

so $\ln S_t \sim \mathcal{N}(\ln S_0 + (\mu - \sigma^2/2)t, \sigma^2 t)$ and:

$$S_t = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right].$$

Theorem 24.3.2: Black-Scholes PDE

Consider a self-financing portfolio in stock S and bond $B = e^{rt}$ that replicates derivative payoff $V(S, t)$. The no-arbitrage condition yields the **Black-Scholes PDE**:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0,$$

with terminal condition $V(S, T) = h(S)$ (payoff function).

Proof

(Delta-hedging derivation.) Let $\Pi = V - \Delta S$ be a hedged portfolio. By Itô's lemma:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \left(\frac{\partial V}{\partial S} - \Delta \right) dS.$$

Setting $\Delta = \partial V / \partial S$ eliminates the stochastic term. The resulting riskless portfolio must earn the risk-free rate: $d\Pi = r\Pi dt$, giving the BS PDE. $\square$

Definition 24.3.3: Risk-Neutral Pricing and the Girsanov Theorem

Under the physical measure $\mathbb{P}$, $dS = \mu S dt + \sigma S dW_t$. Define the market price of risk $\theta = (\mu - r)/\sigma$. By **Girsanov's theorem**, changing measure to $\mathbb{Q}$ via Radon-Nikodým derivative

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\theta W_T - \frac{\theta^2}{2}T\right),$$

the process $\tilde{W}_t = W_t + \theta t$ is a $\mathbb{Q}$ -Brownian motion and:

$$dS = rS dt + \sigma S d\tilde{W}_t.$$

Under $\mathbb{Q}$, the discounted price $e^{-rt}S_t$ is a martingale, and any derivative price is:

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[h(S_T)].$$

For a European call, this integral yields the BSM formula.

Theorem 24.3.4: BSM Formula (Derivation Sketch)

Under $\mathbb{Q}$: $S_T = S_0 \exp[(r - \sigma^2/2)T + \sigma \tilde{W}_T]$, $\tilde{W}_T \sim \mathcal{N}(0, T)$.

$$\begin{aligned} C &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)] \\ &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K) \mathbf{1}_{S_T > K}] \\ &= S_0 \mathbb{Q}(S_T > K \mid S_0) - K e^{-rT} \mathbb{Q}(S_T > K \mid S_0) \\ &= S_0 N(d_1) - K e^{-rT} N(d_2), \end{aligned}$$

where the two probabilities $N(d_1)$ and $N(d_2)$ are computed by standardising the lognormal S_T under $\mathbb{Q}$ and the S -forward measure respectively.

Part VII

Game Theory

Chapter 25

Normal Form Games

25.1 Strategic Form and Dominance

Definition 25.1.1: Normal Form Game

A game $G = (N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N})$ consists of:

- A finite set of **players** $N = \{1, \dots, n\}$.
- For each player i : a **strategy set** S_i (finite for matrix games).
- **Payoff functions** $u_i : S_1 \times \dots \times S_n \rightarrow \mathbb{R}$.

The strategy profile $s = (s_1, \dots, s_n) \in S = \prod_i S_i$. Notation: s_{-i} denotes all players' strategies except i .

Definition 25.1.2: Dominant Strategy

Strategy s_i^* **strictly dominates** s'_i if $u_i(s_i^*, s_{-i}) > u_i(s'_i, s_{-i})$ for all $s_{-i} \in S_{-i}$. **Iterated Elimination of Strictly Dominated Strategies (IESDS)**: repeatedly remove dominated strategies; the order does not matter; any rationalizable strategy survives IESDS.

Definition 25.1.3: Nash Equilibrium

A strategy profile $s^* \in S$ is a **Nash Equilibrium (NE)** if for all players i and all $s_i \in S_i$:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*).$$

No player can profitably deviate given the others' strategies.

Definition 25.1.4: Mixed Strategy Nash Equilibrium

A **mixed strategy** $\sigma_i \in \Delta(S_i)$ is a probability distribution over S_i . Player i 's expected payoff: $u_i(\sigma) = \sum_{s \in S} \sigma_1(s_1) \dots \sigma_n(s_n) u_i(s)$. In a **mixed strategy NE**, each player is indifferent among all strategies in the support of σ_i^* :

$$u_i(s_i, \sigma_{-i}^*) = \text{const} \quad \forall s_i \in \text{supp}(\sigma_i^*).$$

Theorem 25.1.5: Nash Existence Theorem

Every finite game (finite N , finite S_i) has at least one Nash equilibrium in mixed strategies.

Proof

(Sketch.) Define the best-response correspondence $BR_i(\sigma_{-i}) = \arg \max_{\sigma_i} u_i(\sigma_i, \sigma_{-i})$. The joint $BR(\sigma) = \prod_i BR_i(\sigma_{-i})$ maps the compact convex set $\prod_i \Delta(S_i)$ to itself. Kakutani's fixed-point theorem applies (BR is upper hemicontinuous with non-empty convex values), guaranteeing a fixed point $\sigma^* = BR(\sigma^*)$, which is a NE. $\square$

25.2 Classic Examples**Example 25.2.1**

Prisoner's Dilemma: both players choose $\{C, D\}$; payoff matrix:

	C	D
C	(2, 2)	(0, 3)
D	(3, 0)	(1, 1)

D strictly dominates C for both players. Unique NE: (D, D) with payoffs (1, 1) — Pareto dominated by (C, C) . Illustrates the tension between individual rationality and social efficiency.

Example 25.2.2

Battle of the Sexes: two pure NE (Opera, Opera) and (Football, Football), plus one mixed NE where the mixing probabilities balance payoffs.

Chapter 26

Extensive Form Games

26.1 Sequential Games and Backward Induction

Definition 26.1.1: Extensive Form Game

Represented as a game tree with:

- **Nodes:** decision nodes (assigned to a player) and terminal nodes.
- **Information sets:** sets of nodes a player cannot distinguish.
- **Actions:** edges from a decision node.
- **Payoffs:** at each terminal node.

A **strategy** for player i specifies an action at every information set.

Definition 26.1.2: Subgame Perfect Equilibrium (SPE)

A strategy profile is an SPE if it induces a Nash equilibrium in every **subgame** (subtree starting from a single node with complete information). SPE eliminates non-credible threats. **Backward induction:** solve the game tree from terminal nodes upward, choosing the optimal action at each node — yields the unique SPE in finite games of perfect information.

Example 26.1.1

Stackelberg Duopoly: leader (L) moves first, follower (F) observes and responds. Backward induction: F's best response $q_F^{BR}(q_L)$ is computed first; L maximises profit anticipating F's response. Result: L produces more than Cournot; F produces less. First-mover advantage arises from commitment.

Chapter 27

Repeated Games

27.1 Finitely and Infinitely Repeated Games

Definition 27.1.1: Repeated Game

The stage game G is played at $t = 1, 2, \dots$. Players observe all past actions and discount future payoffs by $\delta \in (0, 1)$ per period. Total payoff: $\sum_{t=0}^{\infty} \delta^t u_i(a^t)$.

Theorem 27.1.2: Backward Induction in Finite Repetition

If the stage game has a unique NE, then in the finitely repeated game the unique SPE is to play the stage NE at every period. Cooperation cannot be sustained by the threat of future punishment (unravelling from the last period).

Theorem 27.1.3: Folk Theorem (Informal)

In an infinitely repeated game, if players are sufficiently patient (high δ), any feasible individually rational payoff vector can be supported as a NE payoff (and under stronger conditions, as an SPE payoff via **grim trigger** or **tit-for-tat** strategies).

Proof

(Grim Trigger example, Prisoner's Dilemma.) Cooperate while the opponent has always cooperated; defect forever otherwise. Cooperation is sustainable if the one-shot gain from defecting is outweighed by the loss of future cooperation:

$$3 + \frac{\delta}{1-\delta} \cdot 1 \leq 2 + \frac{\delta}{1-\delta} \cdot 2 \iff \delta \geq \frac{1}{2}.$$

For $\delta \geq 1/2$, cooperation is a subgame-perfect equilibrium. □

Chapter 28

Bayesian Games and Auctions

28.1 Games with Incomplete Information

Definition 28.1.1: Bayesian Game

A game where each player has a **type** $\theta_i \in \Theta_i$ drawn from a joint distribution $F(\theta)$, privately observed. Player i maximises expected payoff over others' types:

$$\max_{a_i(\theta_i)} \mathbb{E}_{\theta_{-i}} [u_i(a_i, a_{-i}(\theta_{-i}), \theta)].$$

A **Bayesian Nash Equilibrium (BNE)** is a strategy profile $a_i^*(\cdot) : \Theta_i \rightarrow A_i$ such that each player's strategy is a best response to the others' strategies, given beliefs about others' types.

28.2 Auction Theory

Definition 28.2.1: Auction Formats

Format	Mechanism
English (ascending)	Open, price rises until one bidder remains; winner pays final price
Dutch (descending)	Price falls until first bidder accepts; strategically equivalent to FPSB
First-Price Sealed Bid (FPSB)	Bidders submit sealed bids; highest bid wins, pays own bid
Second-Price Sealed Bid (Vickrey)	Highest bid wins, pays second-highest bid

Theorem 28.2.2: Revenue Equivalence Theorem

Suppose bidders have independent private values drawn from a common distribution F on $[0, \bar{v}]$, are risk-neutral, and the auction allocates the object to the highest-value bidder with a zero expected payment for the lowest type. Then **all four standard auction formats yield the same expected revenue to the seller.**

Definition 28.2.3: FPSB Equilibrium Bid

With n bidders and values $v_i \sim U[0, \bar{v}]$ i.i.d., the symmetric Bayesian Nash equilibrium bid is:

$$b^*(v) = \frac{n-1}{n} v.$$

Each bidder shades their bid below their value; the shading diminishes as n grows.

Definition 28.2.4: Vickrey Auction (SPSB)

In the second-price auction, bidding own value $b_i(v_i) = v_i$ is a **weakly dominant strategy** for every bidder, regardless of the number of bidders or others' strategies. Truth-telling is optimal because:

$$\text{winning at second price} = \max_{j \neq i} b_j \leq v_i \Rightarrow \text{surplus} = v_i - \max_{j \neq i} b_j > 0.$$

Overbidding risks winning at a price exceeding value; underbidding risks losing a profitable auction.

Remark.

The Myerson optimal auction maximises expected revenue by introducing a **reserve price** r^* : the seller withholds the good if the highest bid falls below r^* . With n bidders and $v_i \sim U[0, 1]$: $r^* = 1/2$ regardless of n .